Linear Regression

From Least Squares to Regularisation

Lebnet Tech Fellows Program

Roadmap

The Problem

Why Squared Loss?

The OLS Solution

Geometric Interpretation

Statistical Properties

Bias-Variance Tradeoff

Regularisation

Model Evaluation

Summary

What Is Linear Regression?

Given an explanatory variable x and a response y, assume:

$$Y = a + bX + \varepsilon$$

where ε captures noise we do not fully understand.

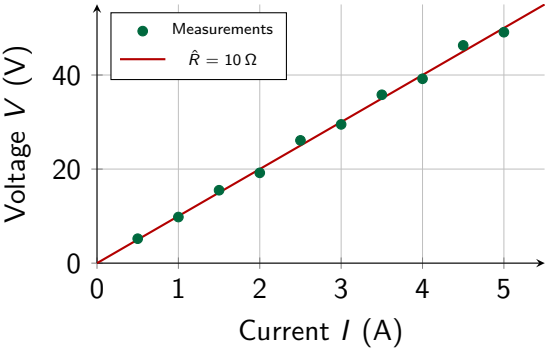
Goal: find estimators $\hat{a}, \hat{b}$ of the true parameters.

- $x \in \mathbb{R}$: univariate regression
- $x \in \mathbb{R}^d$: multivariate regression

Running Example: Ohm's Law

Voltage and current: $V = IR$.

- Measurements (I_i, V_i) from a circuit
- Fit a line to estimate resistance R
- I is explanatory, V is response



Measuring the Error

For a candidate line $\hat{y} = a + bx$, the **vertical distance** from (x_i, y_i) is natural: we are predicting y from x.

The **squared loss**:

$$\ell(y, \hat{y}) = (y - \hat{y})^2$$

Empirical risk (what we minimise):

$$\hat{\mathcal{R}}_n(f) = \frac{1}{n} \sum_{i=1}^{n} (y_i - f(x_i))^2$$

Bayes Optimality of Squared Loss

Under squared loss, the **Bayes optimal predictor** is the conditional expectation:

$$f_*(x) = \mathbb{E}[Y \mid X = x]$$

The **Bayes risk** (irreducible error):

$$\mathcal{R}^* = \mathbb{E}_X\big[\mathrm{Var}(Y \mid X)\big]$$

Linear regression approximates $\mathbb{E}[Y \mid X]$ with the best linear function. If the true conditional expectation is linear, the model is correctly specified.

Matrix Formulation

Stack all observations:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$$

- $\mathbf{y} \in \mathbb{R}^n$: response vector
- $\mathbf{X} \in \mathbb{R}^{n \times d}$: design matrix (first column is all ones for the intercept)
- $\boldsymbol{\theta} \in \mathbb{R}^d$: parameter vector

Empirical risk:

$$\hat{\mathcal{R}}_n(\boldsymbol{\theta}) = \frac{1}{n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2$$

Deriving the OLS Estimator

Set $\nabla \hat{\mathcal{R}}_n = \mathbf{0}$:

$$\frac{2}{n}\left(\mathbf{X}^\top \mathbf{X}\boldsymbol{\theta} - \mathbf{X}^\top \mathbf{y}\right) = \mathbf{0}$$

This gives the **normal equations**:

$$\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{y}$$

If $\mathbf{X}$ has full column rank ($\text{rank}(\mathbf{X}) = d$):

$$\boxed{\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{y}}$$

OLS as Orthogonal Projection

$\mathbf{y}$ typically does not lie in $\mathrm{Im}(\mathbf{X})$.

OLS finds the closest point in that subspace:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\theta}} = \underbrace{\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top}_{\mathbf{P_X}}\,\mathbf{y}$$

- $\mathbf{P_X}$: projection matrix onto $\mathrm{Im}(\mathbf{X})$
- Residuals $\hat{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}}$ are **orthogonal** to every column of $\mathbf{X}$
- This is exactly the content of the normal equations: $\mathbf{X}^\top\hat{\varepsilon} = \mathbf{0}$

Classical Assumptions

Classical Linear Model

1. **Linearity:** $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \varepsilon$
2. **Fixed design:** $\mathbf{X}$ is non-random
3. **Exogeneity:** $\mathbb{E}[\varepsilon \mid \mathbf{X}] = \mathbf{0}$
4. **Spherical errors:** $\mathrm{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$
5. **Full rank:** $\mathrm{rank}(\mathbf{X}) = d < n$

For inference, add:

6. **Normality:** $\varepsilon \mid \mathbf{X} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$

Unbiasedness and Variance

Substitute the true model into OLS:

$$\hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon$$

Under assumptions 1-4

$$\mathbb{E}[\hat{\boldsymbol{\theta}}] = \boldsymbol{\theta} \qquad \text{(unbiased)}$$

$$\mathrm{Var}(\hat{\boldsymbol{\theta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$$

Gauss-Markov: OLS is BLUE

Gauss-Markov Theorem (assumptions 1-5)

Among all **linear unbiased** estimators $\tilde{\boldsymbol{\theta}} = \mathbf{Ay}$:

$$\mathrm{Var}(\tilde{\theta}_j) \geq \mathrm{Var}(\hat{\theta}_j) \quad \text{for all } j$$

Key idea: write $\mathbf{A} = \mathbf{A}_{\mathrm{OLS}} + \mathbf{D}$. Unbiasedness forces $\mathbf{DX} = \mathbf{0}$, which makes the cross terms vanish, leaving $\mathrm{Var}(\tilde{\boldsymbol{\theta}}) = \mathrm{Var}(\hat{\boldsymbol{\theta}}) + \sigma^2 \mathbf{DD}^\top$.

Under normality, OLS is in fact **BUE** (best among *all* unbiased estimators).

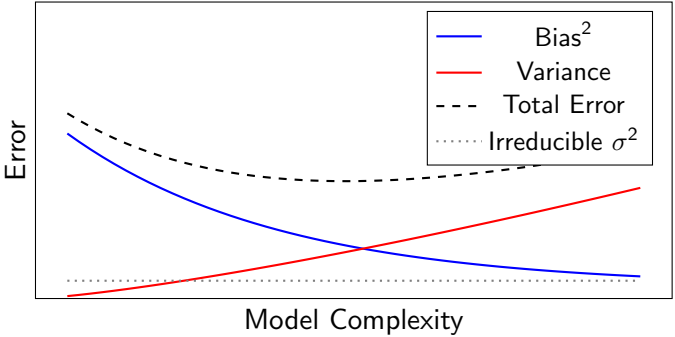
MSE = Bias^2 + Variance

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta})$$

For prediction at a new point $\mathbf{x}_0$:

$$\mathbb{E}[(y_0 - \hat{y}_0)^2] = \underbrace{\sigma^2}_{\text{irreducible}} + \underbrace{\text{Bias}(\hat{y}_0)^2}_{\text{model error}} + \underbrace{\text{Var}(\hat{y}_0)}_{\text{estimation noise}}$$

The Tradeoff in Practice



- **Underfitting** (left): high bias, low variance
- **Overfitting** (right): low bias, high variance
- **Sweet spot**: minimise total error

Ridge Regression (L_2 Penalty)

Add a penalty on $\|\boldsymbol{\theta}\|_2^2$:

Ridge Estimator

$$\hat{\boldsymbol{\theta}}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

- **Biased**, but lower variance than OLS
- Always invertible for $\lambda > 0$ (cures multicollinearity)
- Bayesian view: Gaussian prior $\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$ with $\lambda = \sigma^2/\tau^2$

Lasso (L_1 Penalty)

Replace the L_2 penalty with L_1:

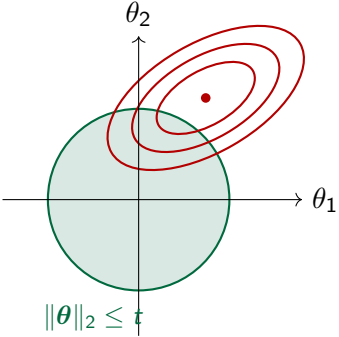
Lasso Estimator

$$\hat{\boldsymbol{\theta}}_{\text{lasso}} = \arg\min_{\boldsymbol{\theta}} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_1 \right\}$$

- No closed-form solution (solved by coordinate descent)
- Produces **exact zeros**: automatic variable selection
- The L_1 ball has corners on the axes; the OLS contour typically first touches a corner

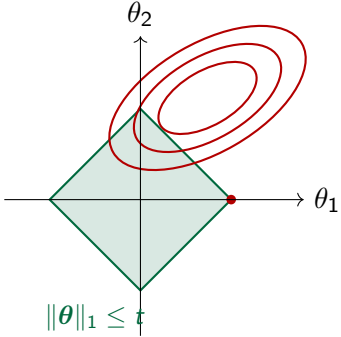
Ridge vs Lasso: Geometry

Ridge (L_2)



All $\theta_j \neq 0$ (shrinkage)

Lasso (L_1)



Some $\theta_j = 0$ (sparsity)

R^2: Proportion of Variance Explained

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

- $R^2 = 1$: perfect fit
- $R^2 = 0$: no better than predicting $\bar{y}$

Problem

R^2 **never decreases** when adding features, even noise.

Fix: Adjusted R^2: $\bar{R}^2 = 1 - \frac{n-1}{n-d}(1 - R^2)$

Cross-Validation

k-fold CV estimates out-of-sample error:

1. Split data into k folds
2. For each fold i: train on all folds except i, evaluate on fold i
3. Average: $\mathrm{CV}_k = \frac{1}{k}\sum_{i=1}^{k} \mathrm{MSE}_i$

- $k = n$ (leave-one-out): low bias, high variance
- $k = 5$ or 10: good bias-variance balance in practice
- Use CV to choose λ in Ridge / Lasso

Recap: The Full Story

1. **Problem:** predict a continuous y from features $\mathbf{x}$.
2. **Loss:** squared error; Bayes optimal predictor is $\mathbb{E}[Y \mid X]$.
3. **OLS:** $\hat{\theta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ (projection onto column space).
4. **Properties:** unbiased, BLUE (Gauss-Markov).
5. **Tradeoff:** MSE $=$ Bias^2 $+$ Variance; complexity must be balanced.
6. **Regularisation:** Ridge (shrinkage), Lasso (sparsity).
7. **Evaluation:** adjusted R^2, cross-validation.

Questions?